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MATH 246A

Homework 4
Due: TBD

Problem 7.1

Suppose $\sum_{n=0}^{\infty} a_n(z-z_0)^n$ is a power series with nonzero coefficients. Show that its radius of convergence R is given by

$$R = \lim_{n\to\infty} \left| \frac{a_n}{a_{n+1}} \right|$$

if the limit exists.

> *Solution.*

Problem 7.2

Find the radius of convergence of the power series $\sum_{n=1}^{\infty} a_n z^n$, where
(a) $a_n = (2i)^n/(n+i)^2$,
(b) $a_n = n^2 + 5in + 1$,
(c) $a_n = n^{\log n}$.

> *Solution.*

Problem 7.3

Show that the series $\sum_{n=1}^{\infty} \frac{z^n}{n}$ converges for $z \in \mathbb{C}$ if and only if $|z| \leq 1$ and $z \neq 1$.

> *Solution.*

Problem 7.4

Suppose that the power series $f(z) = \sum_{n=0}^{\infty} a_n z^n$ and $g(z) = \sum_{n=0}^{\infty} b_n z^n$ have radii of convergence $R_1 > 0$ and $R_2 > 0$, respectively. For $n \in \mathbb{N}_0$ define

$$c_n = \sum_{k=0}^{n} a_k b_{n-k}.$$

Show that the power series $h(z) = \sum_{n=0}^{\infty} c_n z^n$ (the so-called *Cauchy product* of f and g) converges for all $|z| < R := \min\{R_1, R_2\}$ and that $h(z) = f(z)g(z)$ for all such z.

> *Solution.*

Problem 7.5

Let $R > 0$, and suppose that the power series $f(z) = \sum_{n=0}^{\infty} a_n z^n$ converges for all $z \in B := B(0, R)$.
(a) Show that the power series converges uniformly on each compact subset of B.

(b) For $n \in \mathbb{N}_0$ and $z \in B(0,R)$ define

$$S_n(z) := \left| \sum_{k=0}^{n} a_k z^k \right|^2.$$

Show that on each compact subset of B the function sequence $\{S_n\}$ converges uniformly to $|f|^2$.
(c) Show that for each $0 \le r < R$ *Parseval's formula* is valid:

$$\frac{1}{2\pi} \int_0^{2\pi} \left| f(re^{it}) \right|^2 \mathrm{d}t = \sum_{n=0}^{\infty} |a_n|^2 r^{2n}.$$

Solution.

Problem 7.6

(a) Show that the power series

$$\sum_{n=0}^{\infty} \frac{i^{n+1}}{(n+1)!} z^n$$

converges for all $z \in \mathbb{C}$.
(b) Let g be the function represented by the power series in (a). Find a simple expression for $g(z)$ if $z \neq 0$.
(c) For $r > 0$ let $\alpha_r : [0,2] \to \mathbb{C}$ be the path defined by

$$\alpha_r(t) = \begin{cases} re^{i\pi t} & t \in [0,1] \\ (2t-3)rt & \in [1,2]. \end{cases}$$

Show that $\int_{\alpha_r} g(z)\,\mathrm{d}z = 0$. *Hint:* This follows from a general theorem that will be proved later, but show this directly.
(d) Show that $\lim_{r\to\infty} \int_{[-r,r]} g(z)\,\mathrm{d}z = i\pi$.
(e) Show that $\int_0^{\infty} \frac{\sin x}{x}\,\mathrm{d}x = \frac{\pi}{2}$.

Solution.

Problem 7.7

Let $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$.
(a) Show that for each $t \in [0, 2\pi]$ the function P_t defined as

$$P_t(z) = \frac{e^{it} + z}{e^{it} - z} \quad \text{for } z \in \mathbb{D}$$

can be represented by a power series that is centered at $z_0 = 0$ and converges on $\mathbb{D}$.
(b) Let $u : \partial\mathbb{D} \to \mathbb{R}$ be a continuous function. Show that the function $f : \mathbb{D} \to \mathbb{C}$ defined as

$$f(z) = \frac{1}{2\pi} \int_0^{2\pi} \frac{e^{it} + z}{e^{it} - z} u(e^{it})\,\mathrm{d}t \quad \text{for } z \in \mathbb{D}$$

is holomorphic. *Hint:* Find a power series representation of f on $\mathbb{D}$.

> *Solution.*

Problem 7.8

Define $\gamma : [0, 2\pi] \to \mathbb{C}$ as $\gamma(t) = e^{it}$ for $t \in [0, 2\pi]$.
(a) Show that

$$\frac{1}{2\pi i}\int_\gamma \left(w - \frac{1}{w}\right)^n \frac{\mathrm{d}w}{w} = \begin{cases} 0 & n \in \mathbb{N}_0 \text{ odd} \\ (-1)^{n/2}\binom{n}{n/2} & n \in \mathbb{N}_0 \text{ even.} \end{cases}$$

The *Bessel function* $J_0 : \mathbb{C} \to \mathbb{C}$ is defined as

$$J_0(z) = \frac{1}{2\pi i}\int_\gamma \exp\left(\frac{z}{2}\left(w - \frac{1}{w}\right)\right)\frac{\mathrm{d}w}{w}$$

for $z \in \mathbb{C}$.
(b) Show that J_0 is represented by the power series

$$J_0(z) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2}\left(\frac{z}{2}\right)^{2n}$$

convergent for all $z \in \mathbb{C}$ and is hence an entire function, that is, holomorphic on $\mathbb{C}$.

> *Solution.*

Problem 7.9

(a) Show that the function J_0 defined in Problem 7.8 satisfies *Bessel's differential equation*

$$zJ_0''(z) + J_0'(z) + zJ_0(z) = 0$$

for $z \in \mathbb{C}$.
(b) Suppose $f(z) = \sum_{n=0}^{\infty} a_n z^n$ is a power series that converges for $z \in \mathbb{C}$ and the function f represented by this power series satisfies Bessel's differential equation. Show that if $f(0) = 1$ and $f'(0) = 0$, then $f = J_0$. *Hint:* Find a recursive relation that the coefficients a_n must satisfy.

> *Solution.*

Problem 7.10

The periodic oscillations of a thin membrane (such as a drum) can be modeled as follows: in rest position the membrane occupies a region U in the x-y plane. Its movement under periodic (undampened) oscillations is described by a function $u = u(x, y, t)$ measuring the (small) deflection of the membrane in the z-direction in a three-dimensional x-y-z coordinate system over the point $(x, y) \in U$ at time t. The motion of the membrane is then governed by the two-dimensional *wave equation*

$$u_{tt} = u_{xx} + u_{yy}$$

with the boundary condition $u = 0$ on ∂U (with a suitable unit of time depending on material constants of the membrane). Here we denote by a subscript differentiation with respect to the variables t, x, y.

(a) Suppose u is a solution of the wave equation of the form $u(x, y, t) = f(t)v(x, y)$ with sufficiently smooth functions f and v, where $f \neq 0$ is a periodic function with $f(0) = 0$. Show that then there exist constants $A \in \mathbb{R}$ and $\lambda > 0$ such that

$$u(x, y, t) = A \sin(\lambda t)v(x, y),$$

where

$$\Delta v + \lambda^2 v = 0.$$

(b) Suppose that the membrane is circular, say $U = \mathbb{D}$. Suppose that $u(x, y, t) = f(t)g(r)$ is a solution of the wave equation, where f is as in (a) and g only depends on the radius r in polar coordinates in the x-y-plane. Show that g then satisfies the differential equation

$$rg''(r) + g'(r) + \lambda^2 rg(r) = 0.$$

(c) Find solutions g of the equation in (b) in terms of λ and the Bessel function J_0 (see Problem 7.9). Which condition on λ guarantees that $u(x, y, t) = f(t)g(r)$ as in (b) for one of these solutions g satisfies the boundary condition $u = 0$ on $\partial\mathbb{D}$?

> *Solution.*